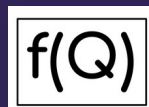


Q-CON 2026 · GDAŃSK · 11 SEPTEMBER

Quantum Neural Networks for Derivatives Pricing

A 2-qubit Quantum Neural Network for Black–Scholes pricing,
benchmarked live across real quantum processors

Sebastian Zajac, PhD



Military University of Technology

Based on arXiv:2604.19832 · in cooperation with Rafał Pracht (finQbit)

11 September 2026 · Gdańsk, Poland · qcon.tech



Small pricing gains, enormous stakes



\$100s T

Notional value of the global OTC derivatives market (BIS, 2023). Much of it carries embedded optionality.



Millions

Valuations run every day for trading, risk, regulatory reporting and stress testing.



Basel III

Rising regulatory demands push for methods that are accurate, scalable and operationally efficient.

Why analysts care: at this scale, a marginal improvement in pricing accuracy or speed translates directly into capital savings and tighter risk control — and a model that learns the pricing map once can then value instantly. That is the prize quantum methods are chasing.

Black–Scholes as a lab, not a destination

The team prices European call options where the exact answer is known — so every error can be measured precisely. The strike is the numeraire: the target is the normalised price $\hat{C} = C / K$ as a function of four inputs.



Moneyness $m = S/K$

0.8 – 1.2



Time to maturity T

0.2 – 1.1 yr



Risk-free rate r

2% – 10%



Volatility σ

1% – 100%

500

training samples

100

out-of-sample test

5

live hardware points

4

input risk factors

The 'ATM convexity challenge'

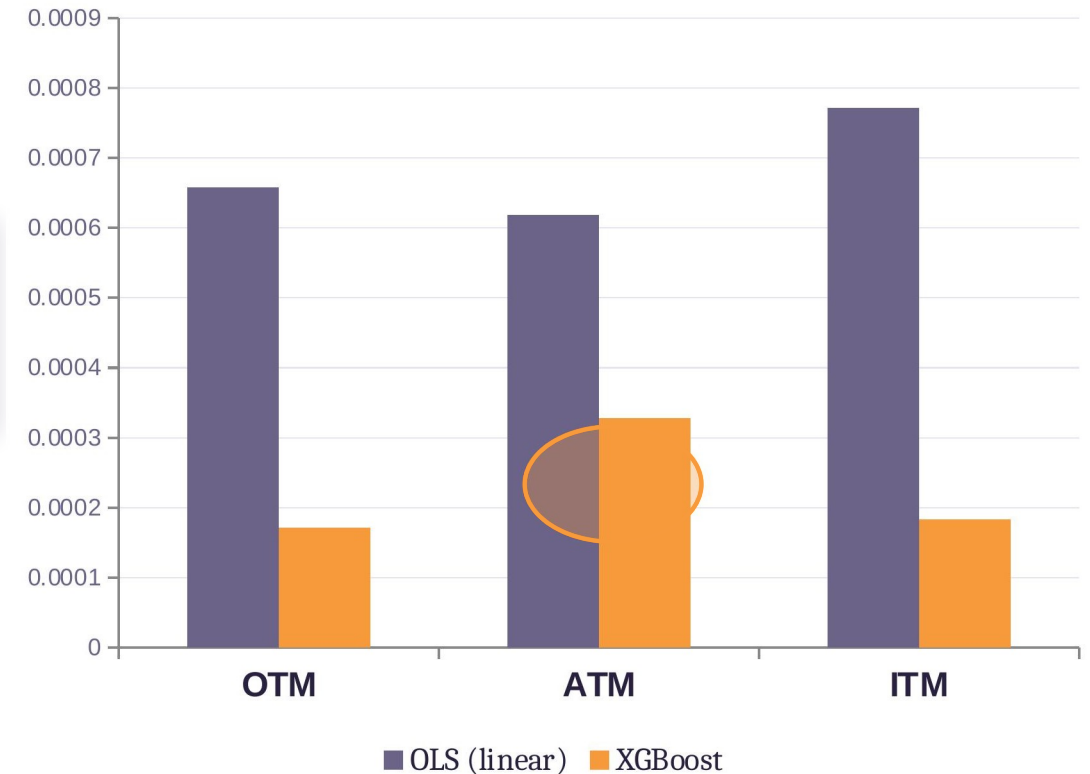
Two classical yardsticks: a linear regression (OLS) and a gradient-boosted tree (XGBoost, ~400 params). Both are weakest exactly at-the-money, where gamma — the curvature of the price surface — peaks.



Tree models use axis-aligned, piecewise-constant steps — **they cannot smoothly trace the sigmoidal ATM transition.**

Highlighted: XGBoost's at-the-money MSE (0.00033) is roughly double its deep OTM / ITM error (~0.00018).

Test-set MSE by moneyness regime · lower is better

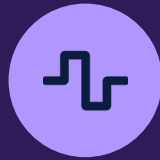


What is a Quantum Neural Network?



Encode

Market inputs (m, T, r, σ) are written into a quantum state as rotation angles — mapping data into a high-dimensional Hilbert space.



Transform

Trainable gates rotate that state. Because the geometry is continuous and smooth, it naturally fits curved surfaces.



Measure

Reading one qubit returns an expectation value — the predicted price. A $\max(0, \cdot)$ step enforces $C \geq 0$.

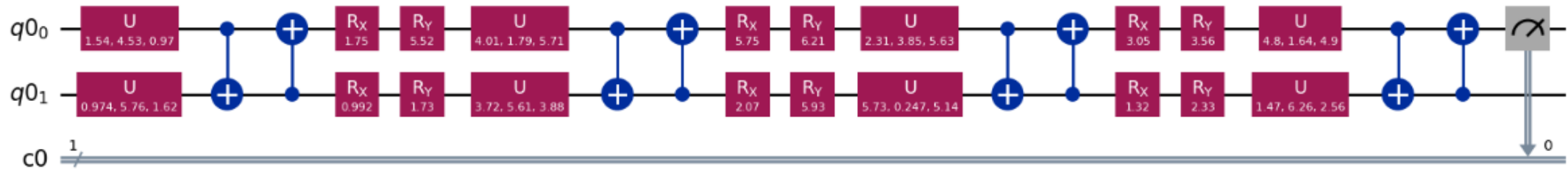


The Fourier view: re-uploading the data several times lets the circuit act like a truncated Fourier series — more layers, more frequencies, so it can resolve the sharp curvature that trips up tree models.

THE MODEL

The finQbit architecture

The finQbit circuit · 2 qubits · 3 re-uploading layers · 36 trainable parameters



Actual transpiled circuit from the paper — dense Ry / Rx encoding, entangling CNOTs, repeated three times, read out on $q0$.



Dense encoding

2 features / qubit (R_y , R_x)



Feature shuffling

pairing permuted per layer



Re-uploading

inputs re-injected each layer



Parameter scaling

learnable weights $w \cdot x$ on inputs



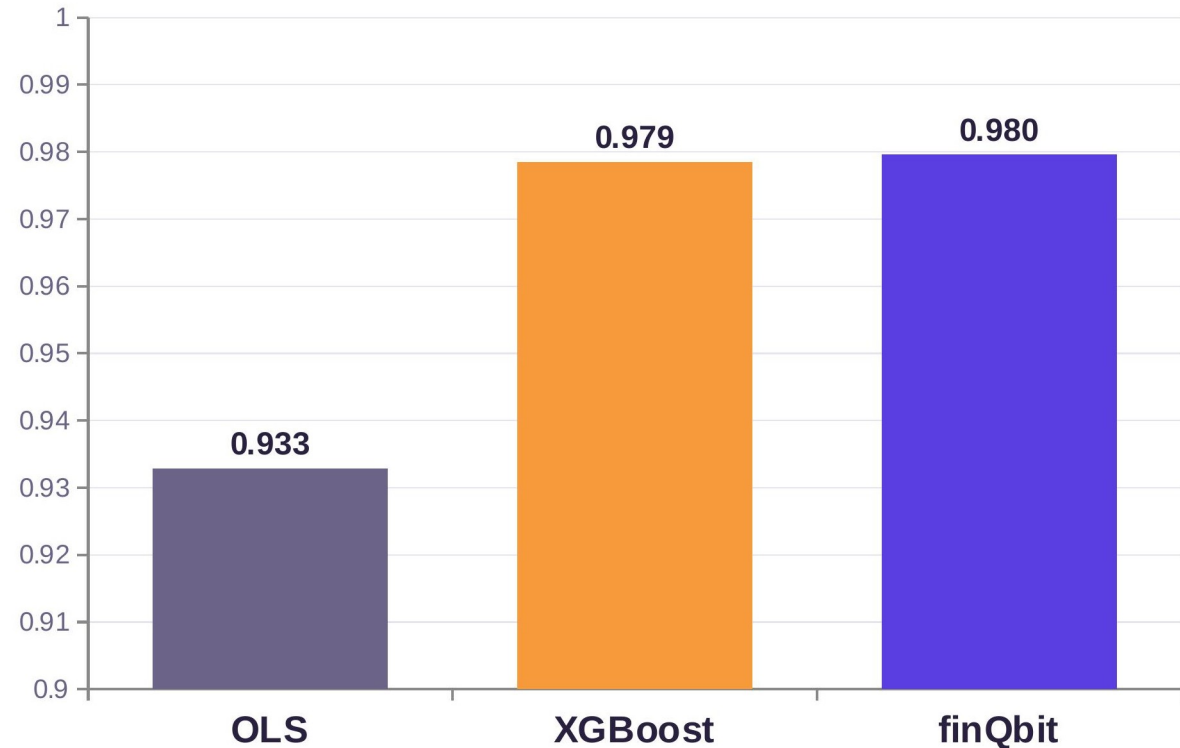
On real hardware: the 2-qubit unitary is compiled via a KAK (Cartan) decomposition into a minimal native-gate circuit — ≤ 3 CNOTs + local rotations — cutting depth and noise.

IBM run only

RESULT 1 — IN SIMULATION

Matches the best tree model, at 1/10th the size

Out-of-sample R^2 · higher is better



Global test metrics (N = 100)

Model	MSE	RMSE	R ²
OLS	0.00069	0.0262	0.933
XGBoost	0.00022	0.0148	0.979
finQbit	0.00021	0.0144	0.980

finQbit slightly edges out XGBoost on every metric — while using ~36 vs ~400 parameters. Hilbert-space rotations are a better inductive bias for a smooth pricing surface than discrete tree splits.

RESULT 2 — WHERE IT COUNTS

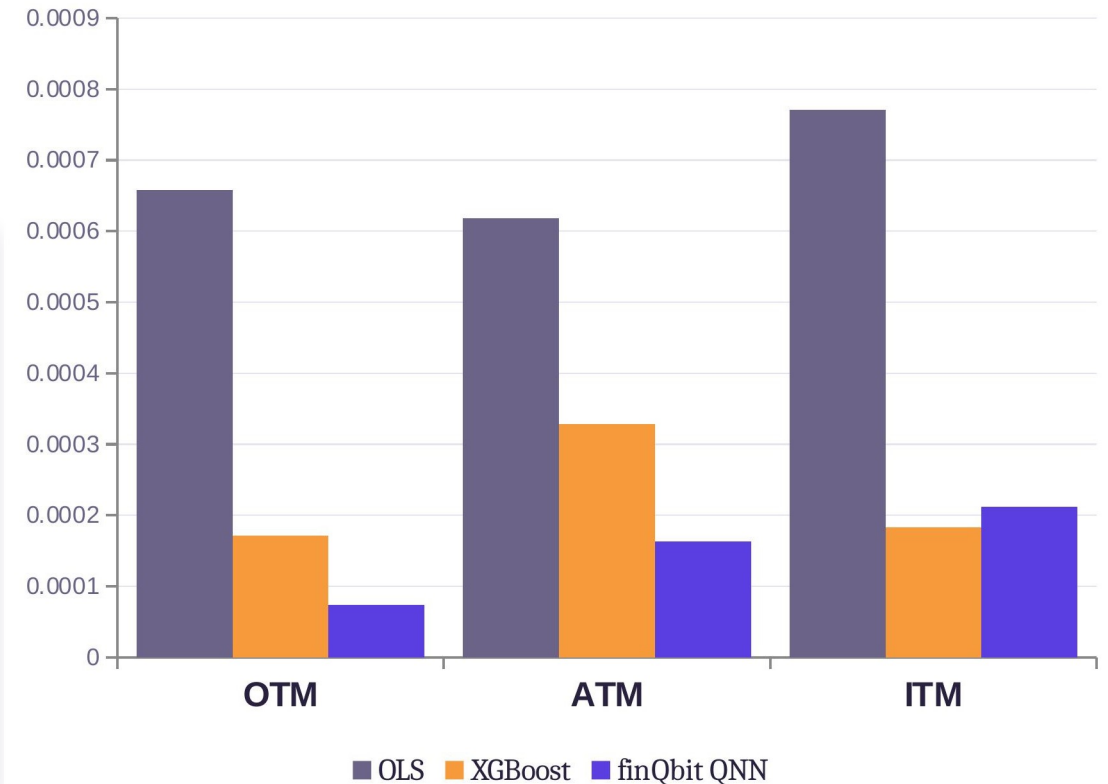
Winning in the high-gamma zone

Adding the QNN to the earlier chart: at-the-money — the hardest region — it roughly halves XGBoost's error. Continuous unitary transforms trace convexity that piecewise-constant trees miss.

~50%

lower MSE than XGBoost in the ATM band ($0.95 \leq m \leq 1.05$), where option gamma peaks.

Test-set MSE by regime · lower is better



The real test: four live quantum computers

The same 2-qubit model was transpiled and executed on four commercial QPUs spanning two qubit technologies — testing whether the results survive real noise, not just idealised simulation.



IBM Fez

SUPERCONDUCTING

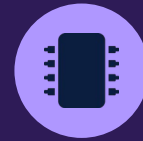
Heavy-hex lattice · nanosecond gates · Qiskit Runtime v2



IonQ Forte

TRAPPED ION

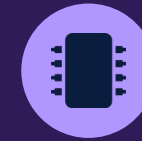
35 Yb⁺ ions · all-to-all connectivity · long coherence



IQM Garnet

SUPERCONDUCTING

20 qubits · square lattice · native CZ gates



Rigetti Ankaa-3

SUPERCONDUCTING

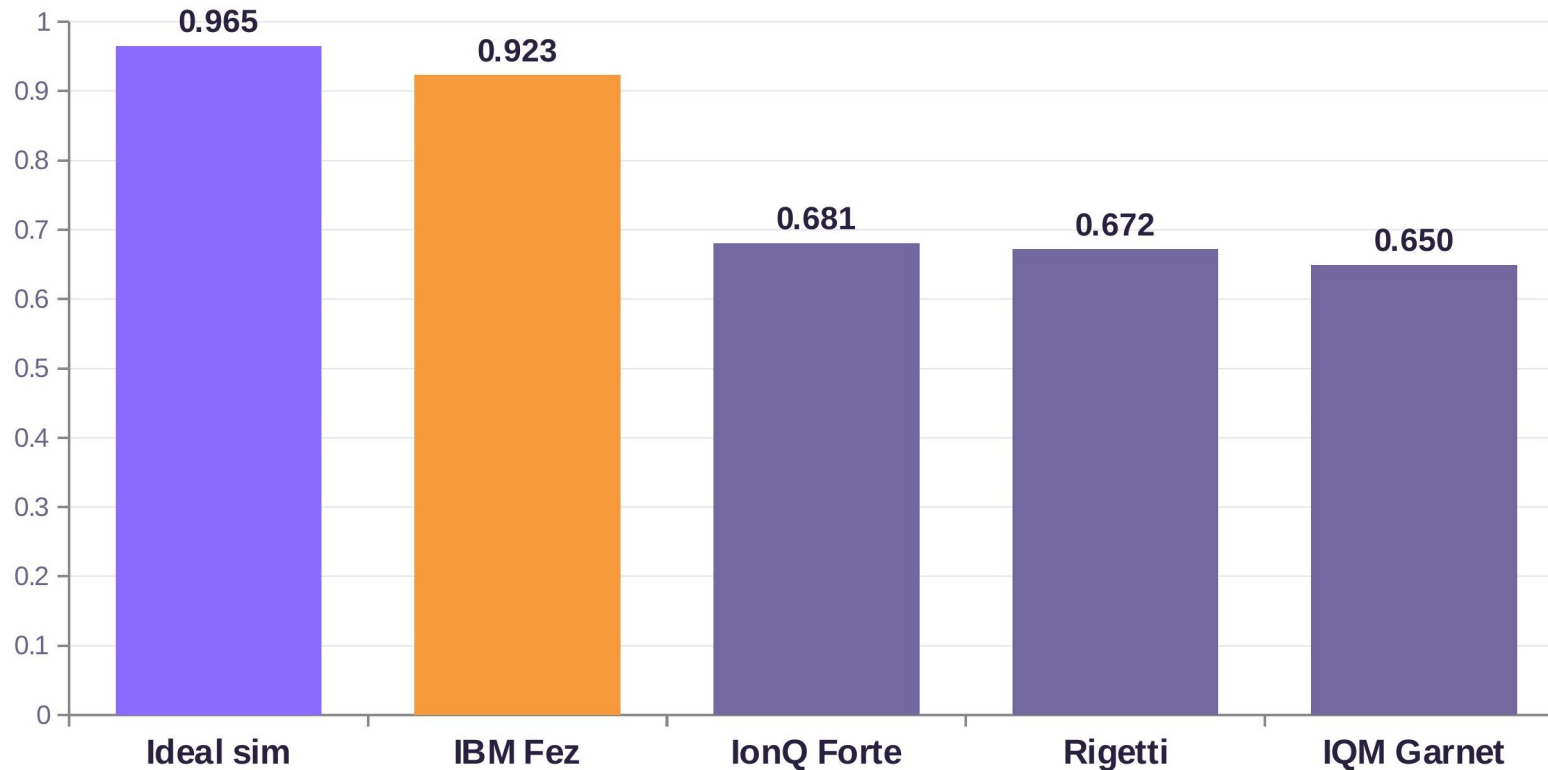
Tunable couplers · octagonal connectivity

Three-tier benchmark: ideal state-vector → shot-noise simulation → full physical QPU execution

CROSS-PLATFORM SCORECARD

How close did real hardware get?

R^2 · R = 20–50 repetitions, N = 2000 shots · higher is better



IBM Fez leads

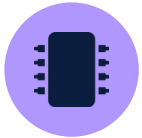
$R^2 > 0.92$ — a superconducting chip nearly reaching ideal-simulation quality on live hardware.



Robust everywhere

Despite different gate sets and topologies, the model held $R^2 > 0.64$ on every device — the architecture travels.

Every machine has a personality



Superconducting → prices too low

IBM, IQM, Rigetti under-price deep in-the-money ($m = 1.2$). Energy relaxation ($|1\rangle \rightarrow |0\rangle$) drains the state before readout — a downward bias.



Trapped ion → prices too high

IonQ is very stable across repetitions but shifts prices upward: systematic gate over-rotation, not random noise, dominates its error.



The mitigation paradox

On IBM Fez and IonQ (readout fidelity $> 99.5\%$), standard readout-error correction made results WORSE — matrix inversion adds more variance than bias it removes.

Takeaway for practitioners: error is systematic and technology-specific — so it can be modelled and corrected, not just averaged away.

How many shots is enough?

QPU time is expensive and queued. The team searched shot counts {500, 2000, 5000} and repetitions {20, 50} to find the point where more sampling stops paying off.



Sweet spot

R = 50

×

2000 shots

Best balance of financial-grade precision and cloud cost.



Diminishing returns follow $1/\sqrt{N}$

Going from 500 → 2000 shots cut the maximum error by ~40%. Beyond ~3000 shots the mean price moves less than 1.5% — extra cost buys almost nothing.



The accuracy-precision paradox (IBM Fez)

More shots ≠ more accurate. Dataset B (50×5000) scored R^2 0.864 vs Dataset A (20×2000) at 0.923 — the longer run drifted out of calibration. Coherence-stability window beats raw sample size.

What an analyst should take away



Targeted edge, not blanket speedup

The win is concentrated where classical models fail — the high-gamma ATM region. Think surgical advantage, not wholesale replacement.



Parameter efficiency is real

10× fewer parameters than XGBoost for equal accuracy suggests lighter models to train, store and audit.



Hardware choice is a modelling choice

Superconducting vs ion-trap bias differ in sign and behaviour. Backend selection and error handling belong in the model design.



Mitigation and shots need tuning

Blind error mitigation can hurt; more shots can drift. Budget and calibration windows are first-class parameters.

What this is — and is not



It IS

- A rigorous proof-of-concept that a tiny QNN can price options on today's noisy hardware.
- Evidence of a structural edge in the high-curvature ATM regime.
- A hardware-agnostic architecture that ran across four platforms unchanged.
- A practical map of sampling cost and hardware bias.



It is NOT

- A replacement for classical pricing of standard European options today.
- A demonstrated quantum speed-up — accuracy, not runtime, is the metric here.
- Tested beyond Black-Scholes; stochastic-vol and multi-asset remain to be shown.
- Free of noise — most devices sit near R^2 0.65 on live hardware.

Beyond the Black–Scholes lab

BSM was the controlled testbed. The next frontier is the models where classical methods actually hurt:



Stochastic volatility

Heston, SABR



Interest-rate dynamics

term-structure models



Multi-asset structures

high-dimensional, path-dependent

The one line to remember: even in the NISQ era, a 2-qubit model already beats classical benchmarks exactly where they fail — a targeted, real advantage worth watching.

Thank you · Questions welcome